

Test 5

Time: 90 minutes

Question 2 (2 points). Verify the following sequents using natural deduction rules.

- (a) $\forall x(P(x) \rightarrow Q(x)), \forall x(Q(x) \rightarrow R(x)) \vdash \forall x(P(x) \rightarrow R(x))$
- (b) $\forall x \forall y(R(x, y) \rightarrow R(y, x)), \exists x \exists y R(x, y) \vdash \exists x \exists y R(y, x)$

Question 3 (2 points). For each of the formulas below, find a model that satisfies the formula and one that does not.

- (a) $\forall x \exists y P(x, y)$
- (b) $\exists x(P(x) \wedge \neg Q(x)) \rightarrow \exists x \neg Q(x)$

Question 4 (2 points). Which of the following semantic entailments hold?

- (a) $\forall x(P(x) \vee Q(x)) \models \forall x P(x) \vee \forall x Q(x)$
- (b) $\forall x \forall y(P(x, y) \rightarrow P(y, x)) \models \forall x \forall y(P(x, y) \leftrightarrow P(y, x))$

Question 5 (2 points). Use the tableau algorithm to determine whether the following formulas are valid or not. If not valid, is it satisfiable?

- (a) $\phi = (p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
- (b) $\psi = \exists x P(x) \rightarrow \forall x P(x)$